Mohammad Yasir Feroz Khan

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Research Interests

- Analysis of PDEs and the calculus of variations with applications in biology and physics.
- Continuum modeling of reaction-diffusion, electrostatics, and coarsening dynamics.
- Numerical methods for PDEs and variational problems.

Education

University of California San Diego Sep. 2021 – Expected Jun. 2027
Ph.D. Candidate in Mathematics
Advisor: Bo Li

University of Cambridge Oct. 2019 – Oct. 2020
Master of Advanced Study in Mathematics
Part III of the Mathematical Tripos

University of Delhi Jul. 2016 – Jun. 2019
B.Sc. (Honours) Mathematics
GPA: 9.66/10

Publications and Manuscripts

Publications and preprints

[1] *Maximizing Reaction via Spatial Localization: A PDE-Constrained Variational Approach* (with B. Li). arXiv:2610.19580 (2026).
Revised and resubmitted to SIAM Journal on Mathematical Analysis.

[2] *Maximizing Reaction via Spatial Localization: Concentration, Concavity, and Absence of Critical Points* (with B. Li). Preprint, 2026.

[3] *Instability of Lagrange Interpolation on* $H^1(\Omega) \cap C(\overline{\Omega})$ (with R. Biswas and R. C. Sau). The Mathematics Student, 88(1–2), 153–158, 2019.

[4] *Beyond Riemann With Volterra, Henstock, and Kurzweil*. Resonance, 24(4), 477–489, 2019.

Manuscripts in preparation

[5] *Regularized Reaction Maximization: Localization and Entropic Spreading* (with B. Li).

[6] *Electrostatic Free Energy with Concentration-Dependent Dielectrics* (with B. Li).

Talks

- *Maximizing Reaction via Spatial Localization.* Contributed talk, Southern California Applied Mathematics Symposium (SOCAMS), Loyola Marymount University, Los Angeles, April 25, 2026.
- *Some PDE Constrained Variational Problems.* Advancement to Ph.D. Candidacy, University of California San Diego, May 29, 2025.

Teaching Experience

Associate Instructor Aug. – Sep. 2026
University of California San Diego

- Selected for the Summer Graduate Teaching Scholars (SGTS) program and served as instructor of record for MATH 20C: Multivariable Calculus, a five-week course for 64 students.
- Designed the course policies and exams, delivered 30 hours of lectures, and supervised two teaching assistants.
- Received 100% positive responses on key student evaluation questions.

Teaching Assistant Sep. 2021 – Present
University of California San Diego

MATH 173A: Optimization Methods for Data Science I	Fall 2026
MATH 103B: Modern Algebra II	Winter 2025, Winter and Spring 2026
MATH 183: Statistical Methods	Fall 2025
MATH 103A: Modern Algebra I	Summer Session I 2022–2025, Fall 2023
MATH 170A: Introduction to Numerical Analysis: Linear Algebra	Winter, Spring, Fall 2024, Spring 2025
MATH 10A: Calculus I	Fall 2021, 2022, Spring 2023
MATH 10B: Calculus II	Winter 2023
MATH 20A: Calculus for Science and Engineering	Winter, Spring 2022

Undergraduate Research

Summer Research Fellow May – Jul. 2018
Indian Institute of Science | Supervisor: Thirupathi Gudi
Constructed a counterexample to H^1-stability of finite-element Lagrange interpolation.

Awards and Honors

Summer Graduate Teaching Scholar, University of California San Diego 2026
Selected following nomination by the Department of Mathematics.

James B. Ax Fellowship, University of California San Diego 2021–2025

BP–Clare Hall India Innovation Masters Studentship, University of Cambridge 2019
One of two recipients from India. Full tuition and a living stipend.

Summer Research Fellowship Programme, IASc–INSA–NASI 2018

Academic Excellence Award, Hansraj College, University of Delhi 2017–2019
First rank in the college's Mathematics program for three consecutive years.

Certificates and Training

Specialist in Student-Centered College Teaching, University of California San Diego Aug. 2026

Erdős Institute Data Science Boot Camp, with distinction May – Aug. 2026

Technical Skills

C++, MATLAB, Python (NumPy, pandas, SciPy, scikit-learn, PyTorch), SQL, LaTeX.